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Homework 4

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Make - up

Suppl. Exam

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Question No.	1	2	3	4	5	6	7	8	9	10	Total
MARKS											

Homework - 4

1. $\Rightarrow$ An operator A is said to be positive i.e. $A \geq 0$ when $A^\dagger = A$ and $\langle \psi | A | \psi \rangle$ is real and non-negative $\forall |\psi\rangle$.

$\Rightarrow \forall \langle \psi | A | \psi \rangle \geq 0 \forall |\psi\rangle$ then $A^\dagger = A$.

$\Rightarrow A = B + iC, B^\dagger = B, C^\dagger = C$

(a) $\Rightarrow \langle \psi | A | \psi \rangle \in \mathbb{R}$.

$\Rightarrow \langle \psi | A | \psi \rangle = \langle \psi | (B + iC) | \psi \rangle$

$= \langle \psi | B | \psi \rangle + i \langle \psi | C | \psi \rangle$

Purely imaginary.

$[\langle \psi | B | \psi \rangle \in \mathbb{R}, \langle \psi | C | \psi \rangle \in \mathbb{R} \text{ as } B \text{ and } C \text{ are Hermitian.}]$

$\Rightarrow \langle \psi | C | \psi \rangle = 0 \rightarrow \langle \psi | iC | \psi \rangle = 0 \forall |\psi\rangle$

$\Rightarrow \langle \psi | B | \psi \rangle \in \mathbb{R}$

$\Rightarrow$ Given B is Hermitian.

$$(\langle \psi | B | \psi \rangle)^* = \langle \psi | B^\dagger | \psi \rangle = \langle \psi | B | \psi \rangle$$

$$\langle \psi | B | \psi \rangle \in \mathbb{R}.$$

$\Rightarrow$ Given C is Hermitian. $\langle \psi | C | \psi \rangle \in \mathbb{R}.$

$\Rightarrow$ Given $\langle \psi | A | \psi \rangle \in \mathbb{R}.$

$$\langle \psi | B | \psi \rangle + i \langle \psi | C | \psi \rangle \in \mathbb{R}$$

$$\boxed{\langle \psi | C | \psi \rangle = 0}$$

$$\boxed{\langle \psi | B | \psi \rangle \in \mathbb{R}}$$

(b) $\Rightarrow C$ is Hermitian; $\langle \psi | C | \psi \rangle = 0 \quad \forall |\psi\rangle.$

$\Rightarrow$ As $C^\dagger = C$, by spectral theorem, it guarantees it has an orthonormal basis of eigenvectors $\{|\phi_k\rangle\}$ with real eigenvalues λ_k .

$$C = \sum_k \lambda_k |\phi_k\rangle \langle \phi_k|, \quad C|\phi_k\rangle = \lambda_k |\phi_k\rangle$$

$\hookrightarrow C$ is unitarily diagonalizable.

Take $|\psi\rangle = |\phi_k\rangle$

$$0 = \langle \phi_k | C | \phi_k \rangle = \langle \phi_k | (\lambda_k |\phi_k\rangle)$$

$$= \lambda_k \langle \phi_k | \phi_k \rangle = \lambda_k \quad (\text{orthonormality})$$

$$\lambda_k = 0 \quad \forall k. \quad \boxed{\text{All eigenvalues of } C \text{ are } 0.}$$

$$\Rightarrow C = \sum_k \lambda_k |\phi_k\rangle \langle \phi_k| = \sum_k 0 \cdot |\phi_k\rangle \langle \phi_k|$$

$$\boxed{C = 0}$$

$$\Rightarrow A = B + iC.$$

$$\boxed{A = B \text{ is Hermitian}}$$

2. $\Rightarrow$ The Hilbert-Schmidt inner product b/w operators A and B is defined as $\langle A, B \rangle = \text{Tr}(A^\dagger B)$.
This is inner product on vector space of operators.
 $\langle A, B \rangle, \langle B, A \rangle$ are conjugates of each other.

(a) $\Rightarrow \langle A, A \rangle = \text{Tr}(A^\dagger A)$.
 $\Rightarrow \forall |\psi\rangle, \langle \psi | A^\dagger A | \psi \rangle = \langle A\psi | A\psi \rangle = \|A\psi\|^2 \geq 0$

$$(A^\dagger A)^\dagger = A^\dagger A$$

$A^\dagger A$ is positive. $A^\dagger A \geq 0$.

All eigenvalues of $A^\dagger A$ are non-negative.

$\text{Tr}(A^\dagger A) = \text{sum of all eigenvalues of } A^\dagger A$.

$$\text{Tr}(A^\dagger A) \geq 0 \longrightarrow \boxed{\langle A, A \rangle \geq 0}$$

(b) $\Rightarrow I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}, Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$

$$\Rightarrow I^\dagger = I, X^\dagger = X, Y^\dagger = Y, Z^\dagger = Z$$

$$\Rightarrow \langle I, X \rangle = \text{Tr}(I^\dagger X) = \text{Tr}(X) = 0$$

$$\textcircled{1} \langle X, I \rangle = 0$$

$$\textcircled{2} \langle I, Y \rangle = \text{Tr}(I^\dagger Y) = \text{Tr}(Y) = 0 = \langle Y, I \rangle$$

$$\textcircled{3} \langle I, Z \rangle = \text{Tr}(I^\dagger Z) = \text{Tr}(Z) = 0 = \langle Z, I \rangle$$

$$\textcircled{4} \langle X, Y \rangle = \text{Tr}(X^\dagger Y)$$

$$\textcircled{5} \langle X, Y \rangle = \text{Tr}(XY)$$

$\Rightarrow$ let P, Q be distinct non-identity Pauli matrices.

$$PQ = -QP$$

$$\begin{aligned} \langle P, Q \rangle &= \text{Tr}(P^\dagger Q) = \text{Tr}(PQ) = \text{Tr}(-QP) \\ &= -\text{Tr}(QP) = -\text{Tr}(PQ) \end{aligned}$$

$$\text{Tr}(PQ) = 0 \quad \langle P, Q \rangle = 0$$

$$\Rightarrow \textcircled{4} \langle X, Y \rangle = \langle Y, X \rangle = 0$$

$$\textcircled{5} \langle X, Z \rangle = \langle Z, X \rangle = 0$$

$$\textcircled{6} \langle Y, Z \rangle = \langle Z, Y \rangle = 0$$

(c) $\Rightarrow$ Any 2×2 matrix A can be written as

$$A = c_1 I + c_2 X + c_3 Y + c_4 Z$$

for some complex scalars c_1, c_2, c_3, c_4 .

$$\Rightarrow \text{Tr}(A) = c_1 \text{Tr}(I) + c_2 \text{Tr}(X) + c_3 \text{Tr}(Y) + c_4 \text{Tr}(Z)$$

[We will use $\text{Tr}(\alpha A + \beta B) = \alpha \text{Tr}(A) + \beta \text{Tr}(B)$]

$$\text{Tr}(A) = 2c_1 + 0 + 0 + 0 \rightarrow \boxed{c_1 = \frac{\text{Tr}(A)}{2}}$$

$$\begin{aligned} \Rightarrow \langle X, A \rangle &= \text{Tr}(X^\dagger A) = \text{Tr}(XA) \\ &= \text{Tr}(c_1 X + c_2 X^2 + c_3 XY + c_4 XZ) \\ &= c_1 \text{Tr}(X) + c_2 \text{Tr}(X^2) + c_3 \text{Tr}(XY) + c_4 \text{Tr}(XZ) \\ &= 0 + 2c_2 + 0 + 0 \rightarrow \boxed{c_2 = \frac{\langle X, A \rangle}{2}} \end{aligned}$$

$\Rightarrow$ Similarly,

$$\begin{aligned} \langle Y, A \rangle &= \text{Tr}(YA) \\ &= \text{Tr}(c_1 Y + c_2 YX + c_3 Y^2 + c_4 YZ) \\ &= c_3 \text{Tr}(Y^2) = 2c_3 \rightarrow \boxed{c_3 = \frac{\langle Y, A \rangle}{2}} \end{aligned}$$

$$\begin{aligned} \Rightarrow \langle Z, A \rangle &= \text{Tr}(ZA) \\ &= \text{Tr}(c_1 Z + c_2 ZX + c_3 ZY + c_4 Z^2) \\ &= c_4 \text{Tr}(Z^2) = 2c_4 \rightarrow \boxed{c_4 = \frac{\langle Z, A \rangle}{2}} \end{aligned}$$

3. $\Rightarrow$ Bloch ball (generalization of Bloch sphere) is a representation of all qubit states, both pure and mixed. This is unit ball in $\mathbb{R}^3$. Surface of this ball, which is the unit sphere, is called Bloch sphere.

$\Rightarrow$ Consider any state ρ of a 2-level system (qubit). We will map this to a point in unit ball. This will be the Bloch ball representation of ρ .

(a) $\Rightarrow$ Given qubit state ρ . ρ is Hermitian. $\rho^\dagger = \rho$.
 $\rho \geq 0$. $\text{Tr}(\rho) = 1$.

$\Rightarrow$ Pauli matrices:
 $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, $X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, $Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$, $Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$

Hilbert-Schmidt inner product: $\langle A, B \rangle = \text{Tr}(A^\dagger B) = \text{Tr}(AB)$

$$\text{Tr}(I^2) = \text{Tr}(X^2) = \text{Tr}(Y^2) = \text{Tr}(Z^2) = 2$$

$$\text{Tr}(XY) = \text{Tr}(YZ) = \text{Tr}(XZ) = 0$$

$$\text{Tr}(IX) = \text{Tr}(IY) = \text{Tr}(IZ) = 0$$

$\Rightarrow \{I, X, Y, Z\}$ is basis for 4-d real vector space of Hermitian matrices, ρ is Hermitian.

$$\rho = \Delta_i I + \Delta_x X + \Delta_y Y + \Delta_z Z$$

(Existence of $\Delta_i, \Delta_x, \Delta_y, \Delta_z$)

$\Rightarrow$ Now we prove they are unique.

$$\rho = (\Delta_i)_1 I + (\Delta_x)_1 X + (\Delta_y)_1 Y + (\Delta_z)_1 Z$$

$$\rho = (\Delta_i)_2 I + (\Delta_x)_2 X + (\Delta_y)_2 Y + (\Delta_z)_2 Z$$

$$0 = (\Delta_i)_1 - (\Delta_i)_2 I + (\Delta_x)_1 - (\Delta_x)_2 X + (\Delta_y)_1 - (\Delta_y)_2 Y + (\Delta_z)_1 - (\Delta_z)_2 Z$$

— (1)

~~Take trace on both sides.~~

Take inner products on both sides as

$$\langle I, 0 \rangle, \langle X, 0 \rangle, \langle Y, 0 \rangle, \langle Z, 0 \rangle$$

$$(\Delta_i)_1 = (\Delta_i)_2, (\Delta_x)_1 = (\Delta_x)_2, (\Delta_y)_1 = (\Delta_y)_2, (\Delta_z)_1 = (\Delta_z)_2$$

(Unique)

$\Rightarrow$ Given a qubit state ρ , $\exists$ unique real nos $\Delta_i, \Delta_x, \Delta_y, \Delta_z$ such that

$$2\rho = \Delta_i I + \Delta_x X + \Delta_y Y + \Delta_z Z$$

$\Rightarrow$ [Here, from HW3, Question 10(a), every 2×2 Hermitian operator is a linear combination of Pauli operators with real coefficients.]

$$(b) \Rightarrow 2\rho = \Delta_i I + \Delta_x X + \Delta_y Y + \Delta_z Z$$

$$\text{Tr}(2\rho) = \text{Tr}(\Delta_i I + \Delta_x X + \Delta_y Y + \Delta_z Z)$$

$$2\text{Tr}(\rho) = \Delta_i \text{Tr}(I) + 0 + 0 + 0 = 2\Delta_i$$

$$[\because \text{Tr}(X) = \text{Tr}(Y) = \text{Tr}(Z) = 0] [\text{Tr}(\rho) = 1]$$

$$\boxed{\Delta_i = 1}$$

$$(c) \Rightarrow 2\rho = I + \Delta_x X + \Delta_y Y + \Delta_z Z$$

$$\Rightarrow \text{let } A = \Delta_x X + \Delta_y Y + \Delta_z Z$$

$$\Rightarrow A^2 = \Delta_x^2 X^2 + \Delta_y^2 Y^2 + \Delta_z^2 Z^2$$

$$+ 2\Delta_x \Delta_y (XY + YX)$$

$$+ 2\Delta_y \Delta_z (YZ + ZY) + 2\Delta_x \Delta_z (XZ + ZX)$$

$$\Rightarrow X^2 = Y^2 = Z^2 = I$$

$$XY = -YX, YZ = -ZY, XZ = -ZX$$

$$A^2 = (\Delta x^2 + \Delta y^2 + \Delta z^2) \mathbb{I}$$

$\Rightarrow$ If λ is eigenvalue of A , then λ^2 is eigenvalue of A^2

$$\lambda^2 = \Delta x^2 + \Delta y^2 + \Delta z^2$$

$$\lambda = \pm \sqrt{\Delta x^2 + \Delta y^2 + \Delta z^2} = \pm \Delta$$

$$\Rightarrow \rho = \frac{1}{2}(\mathbb{I} + A)$$

$$\lambda_\rho = \frac{1}{2}(1 + \lambda), \quad \lambda = \pm \sqrt{\Delta x^2 + \Delta y^2 + \Delta z^2} = \pm \Delta$$

$$\text{where } \Delta = \sqrt{\Delta x^2 + \Delta y^2 + \Delta z^2}$$

$$\boxed{\lambda_\rho = \frac{1 \pm \Delta}{2}} \rightarrow 2 \text{ eigenvalues (real) of } \rho$$

(d) $\Rightarrow \rho$ is density operator. $\rho \geq 0$. $\Delta = \sqrt{\Delta x^2 + \Delta y^2 + \Delta z^2}$

$\Rightarrow$ Eigenvalues of ρ are non-negative.

$$\Rightarrow \frac{1+\Delta}{2} \geq 0, \quad \frac{1-\Delta}{2} \geq 0 \quad \left. \begin{array}{l} -1 \leq \Delta \leq 1 \\ \Delta = \sqrt{\Delta x^2 + \Delta y^2 + \Delta z^2} \\ \Delta \geq 0 \end{array} \right\}$$

$$\Delta \geq -1, \quad \Delta \leq 1$$

$$\Rightarrow \boxed{0 \leq \Delta \leq 1} \rightarrow \text{allowable range of values for } \Delta$$

(e) $\Rightarrow$ Case $\Delta = 1$:

$$\lambda_\rho = \frac{1 \pm \Delta}{2} = 1, 0 \rightarrow \text{rank}(\rho) = 1$$

$\rightarrow$ pure state

$\Rightarrow$ Case $\Delta < 1$:

$$\lambda_\rho = \frac{1 \pm \Delta}{2} \Rightarrow \lambda_1 = \frac{1+\Delta}{2}, \quad \lambda_2 = \frac{1-\Delta}{2}$$

$$0 < \lambda_1 < 1, \quad 0 < \lambda_2 < 1$$

$$\text{rank}(\rho) = 2 \longrightarrow \boxed{\text{mixed state}}$$

6. $\Rightarrow$ Consider any d -level quantum system

$$\Rightarrow \text{Purity of state } \rho = \text{Tr}(\rho^2)$$

$\Rightarrow$ Take any density operator ρ .

$$\rho^\dagger = \rho, \quad \rho \geq 0, \quad \text{Tr}(\rho) = 1.$$

$$\rho = \sum_{i=1}^d p_i |i\rangle\langle i| \quad \{ |i\rangle \} \text{ is orthonormal basis, } p_i \geq 0.$$

$$\sum_{i=1}^d p_i = 1.$$

$$\Rightarrow \rho^2 = \left(\sum_{i=1}^d p_i |i\rangle\langle i| \right) \left(\sum_{j=1}^d p_j |j\rangle\langle j| \right)$$

$$= \sum_{i=1}^d \sum_{j=1}^d p_i p_j |i\rangle\langle i|j\rangle\langle j|$$

$$= \sum_{i=1}^d \sum_{j=1}^d p_i p_j |i\rangle\langle j| \delta_{ij}$$

$$\rho^2 = \sum_{i=1}^d p_i^2 |i\rangle\langle i|$$

$$\text{Tr}(\rho^2) = \text{Tr} \left(\sum_{i=1}^d p_i^2 |i\rangle\langle i| \right) = \sum_{i=1}^d p_i^2$$

(a) $\Rightarrow p_i \geq 0, \quad \sum_{i=1}^d p_i = 1, \quad \text{Tr}(\rho^2) = \sum_{i=1}^d p_i^2$

$$\Rightarrow \sum_{i=1}^d p_i^2 \leq \left(\sum_{i=1}^d p_i \right)^2$$

$$\Rightarrow \boxed{\text{Tr}(\rho^2) \leq 1}$$

(b) ① $\Rightarrow$ Given ρ is a pure state
 $\Rightarrow \rho = |\psi\rangle\langle\psi|$. $\text{rank}(\rho) = 1$.
 Eigenvalues = $1, 0, 0, \dots, 0$ of ρ
 $\Rightarrow \text{Tr}(\rho^2) = 1^2 + 0^2 + \dots + 0^2 = 1$ Eigenvalues of ρ^2
 $\Rightarrow \text{Tr}(\rho^2) = 1 + 0 + 0 + \dots + 0 \Rightarrow \boxed{\text{Tr}(\rho^2) = 1}$

② $\Rightarrow$ Given $\text{Tr}(\rho^2) = 1 \rightarrow \sum_{i=1}^d p_i^2 = 1$
 $\Rightarrow$ We know $p_i \geq 0, \sum_{i=1}^d p_i^2 = 1$.
 $\Rightarrow \sum_{i=1}^d p_i^2 \leq \left(\sum_{i=1}^d p_i\right)^2 \Rightarrow \sum_{i=1}^d p_i^2 \leq 1$
 $\hookrightarrow$ equality iff exactly one p_i is non zero

$\Rightarrow$ Eigenvalues of $\rho^2 = 1, 0, 0, \dots, 0$

Eigenvalues of $\rho = 1, 0, 0, \dots, 0$

$\Rightarrow \rho = |\psi\rangle\langle\psi| \rightarrow \boxed{\rho \text{ is pure state}}$

By ① and ②, $\text{Tr}(\rho^2) = 1$ iff ρ is pure.

7. $\Rightarrow$ If ρ is state of a d -level system then
 $\text{Tr}(\rho^2) \leq 1$.

(a) $\Rightarrow$ Take any state ρ of a d -level system.

$\Rightarrow \langle A, B \rangle_{\text{HS}} = \text{Tr}(A^\dagger B)$, $\|A\|_{\text{HS}}^2 = \text{Tr}(A^\dagger A)$

$\Rightarrow \|\rho\|_{\text{HS}}^2 = \text{Tr}(\rho^\dagger \rho) = \text{Tr}(\rho \rho) = \text{Tr}(\rho^2)$

$\|\mathbb{I}\|_{\text{HS}}^2 = \text{Tr}(\mathbb{I}^\dagger \mathbb{I}) = \text{Tr}(\mathbb{I}) = d$

HS
 $\rightarrow$ Hilbert
 Schmidt

$$\Rightarrow \langle \rho, I \rangle_{HS} = \text{Tr}(\rho^\dagger I) = \text{Tr}(\rho) = 1$$

$\Rightarrow$ By Cauchy Schwartz inequality,

$$|\langle \rho, I \rangle_{HS}|^2 \leq \|\rho\|_{HS}^2 \|I\|_{HS}^2$$

$$1 \leq \text{Tr}(\rho^2) \cdot d$$

$$\boxed{\text{Tr}(\rho^2) \geq \frac{1}{d}}$$

(b) $\Rightarrow \text{Tr}(\rho^2) = \frac{1}{d} \rightarrow$ Equality in Cauchy-Schwartz

$\Rightarrow \rho$ and I are linearly independent.

$$\rho = \lambda I \quad \lambda \text{ is a constant.}$$

$$\text{Tr}(\rho) = 1 = \text{Tr}(\lambda I) = \lambda \text{Tr}(I) = \lambda d.$$

$$\lambda = \frac{1}{d}.$$

$\Rightarrow$ Unique state ρ with purity $\frac{1}{d}$:

$$\boxed{\rho = \frac{1}{d} I}$$

$$\text{Tr}(\rho^2) = \frac{1}{d}$$

Only 1 state that satisfies this condition as linear independence of ρ on I and property

$$\text{Tr}(\rho) = 1 \text{ fixes } \lambda \text{ in } \rho = \lambda I.$$

Maximally mixed / completely mixed state

THE END